

Exp No:4a

Data Inspection, Filtering

Date:

AIM

To perform data inspection, filtering, and sorting operations on a social media dataset using Python and Pandas, in order to understand the structure of the dataset and extract meaningful subsets of data based on specified conditions.

ALGORITHM

Step 1: Import the required library

Import the Pandas library for data manipulation and analysis.

Step 2: Load the dataset

Read the social media CSV dataset using the `pandas.read_csv()` function and store it in a DataFrame named `df`.

Step 3: Inspect the dataset

Display the first few records using `df.head()` to understand the structure and contents of the dataset.

Step 4: Check dataset dimensions

Use `df.shape` to determine the number of rows and columns present in the dataset.

Step 5: Check data types

Use `df.dtypes` to identify the data type of each column, such as integer, floating-point, or categorical/string data.

Step 6: Filter records based on shares

Apply a condition to select posts having more than 85 shares using:

Code:

1.

```
import pandas as pd

df = pd.read_csv(
    r"C:\Users\SHREE HARINI\Downloads\social_media_dataset_with_null_duplicates_outliers.csv"
)

print(df.head())
```

	Post_ID	Platform	Likes	Comments	Shares	Views	Sentiment
0	1	Instagram	304.0	384.0	71	17049	Positive
1	2	Facebook	939.0	351.0	190	36741	Positive
2	3	YouTube	250000.0	21.0	8	7140	Positive
3	4	Facebook	4239.0	313.0	7	37781	Positive
4	5	YouTube	3536.0	117.0	115	39618	Neutral

2.

```
print(df.shape)
```

(53, 7)

3.

```
print(df.dtypes)
```

```
Post_ID      int64
Platform     object
Likes        float64
Comments     float64
Shares       int64
Views        int64
Sentiment    object
dtype: object
```

4.

```
most_shared = df[df["Shares"]>85]
print(most_shared)
```

	Post_ID	Platform	Likes	Comments	Shares	Views	Sentiment
1	2	Facebook	939.0	351.0	190	36741	Positive
4	5	YouTube	3536.0	117.0	115	39618	Neutral
5	6	Instagram	NaN	362.0	109	23298	Neutral
6	7	Facebook	1863.0	495.0	196	23059	Positive
7	8	Instagram	3212.0	54.0	92	23541	Negative
8	9	Twitter	455.0	378.0	118	36142	Positive
10	11	Twitter	4829.0	103.0	181	5558	Positive
12	13	NaN	3814.0	330.0	94	11659	Neutral
14	15	Instagram	1501.0	278.0	187	17043	Positive
15	16	LinkedIn	3208.0	143.0	164	46099	Negative
16	17	Facebook	2756.0	436.0	197	4665	Positive
18	19	YouTube	2677.0	113.0	168	1000000	Neutral
20	21	YouTube	4515.0	NaN	192	39311	Neutral
29	30	Facebook	4518.0	493.0	200	35757	Positive
33	34	Facebook	3993.0	489.0	141	11821	NaN
35	36	Facebook	2653.0	209.0	172	43590	Neutral
37	38	Instagram	2869.0	15.0	151	37301	Positive
40	41	YouTube	2049.0	5000.0	172	32812	Positive
41	42	YouTube	1183.0	375.0	147	38762	Neutral
42	43	Facebook	3974.0	418.0	105	13478	Positive
43	44	Instagram	3631.0	186.0	109	27941	Neutral
44	45	Instagram	906.0	36.0	104	48724	Neutral
47	48	Instagram	3730.0	418.0	141	7416	Positive
48	49	YouTube	220.0	52.0	193	16491	Positive
50	9	Twitter	455.0	378.0	118	36142	Positive
51	16	LinkedIn	3208.0	143.0	164	46099	Negative

5.

```
facebook_users=df[df["Platform"]=="Facebook"]
print(facebook_users)
```

	Post_ID	Platform	Likes	Comments	Shares	Views	Sentiment
1	2	Facebook	939.0	351.0	190	36741	Positive
3	4	Facebook	4239.0	313.0	7	37781	Positive
6	7	Facebook	1863.0	495.0	196	23059	Positive
11	12	Facebook	2470.0	45.0	60	7619	Neutral
16	17	Facebook	2756.0	436.0	197	4665	Positive
23	24	Facebook	3558.0	310.0	17	26216	Neutral
29	30	Facebook	4518.0	493.0	200	35757	Positive
33	34	Facebook	3993.0	489.0	141	11821	NaN
35	36	Facebook	2653.0	209.0	172	43590	Neutral
42	43	Facebook	3974.0	418.0	105	13478	Positive
46	47	Facebook	3556.0	98.0	72	31318	Positive

6.

```
df_sorted = df.sort_values(by="Views")
print(df_sorted)
```

	Post_ID	Platform	Likes	Comments	Shares	Views	Sentiment
16	17	Facebook	2756.0	436.0	197	4665	Positive
17	18	Instagram	2684.0	210.0	69	5337	Positive
10	11	Twitter	4829.0	103.0	181	5558	Positive
2	3	YouTube	250000.0	21.0	8	7140	Positive
47	48	Instagram	3730.0	418.0	141	7416	Positive
11	12	Facebook	2470.0	45.0	60	7619	Neutral
22	23	LinkedIn	844.0	391.0	13	8185	Positive
30	31	YouTube	2755.0	255.0	5	8331	Neutral
21	22	YouTube	3371.0	190.0	57	10065	Negative
28	29	Twitter	4258.0	316.0	51	11016	Neutral
12	13	NaN	3814.0	330.0	94	11659	Neutral
33	34	Facebook	3993.0	489.0	141	11821	NaN
42	43	Facebook	3974.0	418.0	105	13478	Positive
48	49	YouTube	220.0	52.0	193	16491	Positive
14	15	Instagram	1501.0	278.0	187	17043	Positive
0	1	Instagram	304.0	384.0	71	17049	Positive
19	20	LinkedIn	1270.0	140.0	36	17162	Negative
36	37	LinkedIn	4339.0	236.0	31	17246	Positive
39	40	Instagram	1975.0	39.0	9	22654	Positive
6	7	Facebook	1863.0	495.0	196	23059	Positive
5	6	Instagram	NaN	362.0	109	23298	Neutral
7	8	Instagram	3212.0	54.0	92	23541	Negative
23	24	Facebook	3558.0	310.0	17	26216	Neutral
49	50	LinkedIn	4078.0	251.0	55	27282	Positive
43	44	Instagram	3631.0	186.0	109	27941	Neutral
26	27	Instagram	2504.0	227.0	41	30735	Positive
46	47	Facebook	3556.0	98.0	72	31318	Positive
40	41	YouTube	2049.0	5000.0	172	32812	Positive
27	28	Twitter	4200.0	395.0	46	34271	Positive
29	30	Facebook	4518.0	493.0	200	35757	Positive
32	33	Instagram	4081.0	422.0	18	35911	Positive
8	9	Twitter	455.0	378.0	118	36142	Positive
50	9	Twitter	455.0	378.0	118	36142	Positive

RESULT: The social media dataset is successfully loaded and inspected. Records satisfying specific conditions such as Shares greater than 85 and Facebook platform posts are filtered successfully. The dataset is also sorted based on the Views column in ascending order.